

# Week 5, Lecture 2 WOD: Answer Key

## Writing the Likelihood

NRES 746, Instructor Answer Key (not for student distribution)

Each item repeats the handout's question, then gives the answer, a short rationale, and mistakes to watch for. Questions with one correct answer are answered directly. Open-ended items (1d, 2d, 2e, the group question, 4c, 5c, and the reflection) get discussion points instead. All numbers are computed in R when this file is knitted.

Data used throughout: four nests with outcomes  $y = (1, 1, 0, 1)$  and sagebrush cover  $x = (30, 20, 10, 30)$  percent.

### Part 1: One observation

1a.

Write down the probability of each possible outcome for a single nest.  $P(y = 1 \mid p) = \underline{\hspace{2cm}}$   
 $P(y = 0 \mid p) = \underline{\hspace{2cm}}$

**Answer.**  $P(y = 1 \mid p) = p$  and  $P(y = 0 \mid p) = 1 - p$ .

**Why.** The two outcomes are the only possibilities, so their probabilities sum to 1.

**Watch for.** Nothing much. This is a warm-up.

1b.

Write a single formula that gives the probability of an observed outcome  $y$ . Make sure the formula gives the correct answer for both possible outcomes of  $y$  (0 or 1).

**Answer.**  $P(y \mid p) = p^y(1 - p)^{1-y}$ . Also correct:  $yp + (1 - y)(1 - p)$ .

**Why.** If  $y = 1$ , the second factor is  $(1 - p)^0 = 1$ , leaving  $p$ . If  $y = 0$ , the first factor is  $p^0 = 1$ , leaving  $1 - p$ . This is the binomial pmf with one trial, since  $\binom{1}{y} = 1$ .

**Watch for.**

- $p^y(1 - p)$ , with the exponent missing on the second factor.
- $yp$  alone, which gives 0 when  $y = 0$ .
- A piecewise “if” rule, which doesn’t answer the question.

1c.

Nest 1 hatched ( $y_1 = 1$ ). Evaluate the likelihood of  $p = 0.75$  and of  $p = 0.25$  given this observation.

**Answer.**  $\mathcal{L}_1(0.75) = 0.75$  and  $\mathcal{L}_1(0.25) = 0.25$ .

**Why.** With  $y = 1$ , the formula from 1b reduces to  $p$ . The value  $p = 0.75$  makes the observed outcome three times as probable as  $p = 0.25$ .

**Watch for.** Plugging  $y$  and  $p$  into the wrong slots, which gives 0.75 for both or reverses them.

1d.

In your formula from 1b, which symbol is the data and which is the parameter?

**Answer.**  $y$  is the data and  $p$  is the parameter.

**Why (discussion).** In a probability calculation (Lab 3),  $p$  was given and we asked about  $y$ . In a likelihood,  $y$  has already been observed and  $p$  is what varies. The formula is the same, and only which symbol is fixed changes. This is the central idea of the WOD, so listen for whether students can say it in their own words.

**Watch for.** Calling  $p$  “the probability” and  $y$  “the variable” without saying which one is fixed.

## Part 2: Several observations

**2a.**

Write the likelihood of all four observations,  $\mathcal{L}_4(p)$ , in terms of your formula from 1b. We can assume all observations are independent! Please use product notation ( $\prod$ ) to write the likelihood.

**Answer.**

$$\mathcal{L}_4(p) = \prod_{i=1}^4 p^{y_i} (1-p)^{1-y_i}$$

**Why.** Independence means the joint probability of all four outcomes is the product of the individual probabilities.

**Watch for.**

- Adding instead of multiplying (confusing “or” with “and”).
- Dropping the subscript  $i$ , so every term looks identical.

**2b.**

Plug in the observed outcomes and simplify to the form  $p^j(1-p)^k$ . What are  $j$  and  $k$ ?

**Answer.**  $j = 3$ ,  $k = 1$ , so  $\mathcal{L}_4(p) = p^3(1-p)$ .

**Why.** Three nests hatched (each contributes  $p$ ) and one failed (contributes  $1-p$ ). The order of the nests doesn’t matter.

**Watch for.** A student who writes  $\binom{4}{3}p^3(1-p)$  has used the binomial with  $N = 4$ . Accept it. The constant 4 scales every value in 2c by 4 but doesn’t move the peak, which is a good point to raise in discussion.

**2c.**

Evaluate your answer to 2b at the three values of  $p$  below. Fractions are fine.

**Answer.**

| $p$  | $\mathcal{L}_4(p)$                    |
|------|---------------------------------------|
| 0.25 | $(1/64)(3/4) = 3/256 \approx 0.012$   |
| 0.50 | $(1/8)(1/2) = 1/16 = 0.0625$          |
| 0.75 | $(27/64)(1/4) = 27/256 \approx 0.105$ |

**Watch for.** Arithmetic slips with  $(3/4)^3 = 27/64$ .

**2d.**

Add up your three likelihood values. Do they sum to 1? Should they?

**Answer.** They sum to  $46/256 \approx 0.18$ . No, and they shouldn’t have to.

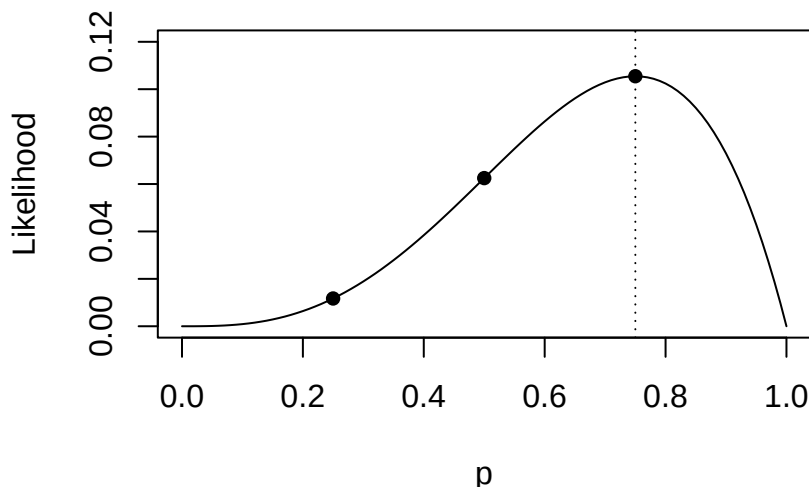
**Why (discussion).** Probabilities sum to 1 across the possible *outcomes* with  $p$  fixed. A likelihood runs across possible *parameter values* with the data fixed, and nothing forces those to add to 1. Even the area under the whole curve from 0 to 1 is only 0.05 (1/20). So  $\mathcal{L}_4(0.75)$  is not “the probability that  $p = 0.75$ .” It is the probability of these data if  $p$  were 0.75.

**Watch for.** Students who think they made an arithmetic error because the sum isn't 1. This question is designed to surface that assumption.

**2e.**

Plot your three values on the axes below, then sketch the whole curve  $\mathcal{L}_4(p)$  for  $0 \leq p \leq 1$ . Mark where you think the peak is. Is that where you would have guessed from the raw data alone?

**Answer.** The curve is 0 at both ends and peaks at  $p = 0.75$ :



**Why.** The peak matches the raw proportion (3 of 4 hatched). Setting the derivative to zero confirms it:  $\frac{d}{dp} p^3(1-p) = p^2(3-4p) = 0$  at  $p = 3/4$ . Students weren't asked for the derivative.

**Watch for.** A symmetric curve peaked at 0.5, or a curve that doesn't go to 0 at  $p = 0$  and  $p = 1$ .

### Part 3: The log-likelihood

**3a.**

Take the log of your product from 2a and use the log rules to rewrite it as a sum. Write your result for a general set of  $n$  nests with outcomes  $y_1, \dots, y_n$ .

**Answer.**

$$\ell_n(p) = \sum_{i=1}^n [y_i \log p + (1 - y_i) \log(1 - p)]$$

Equivalently,  $(\sum y_i) \log p + (n - \sum y_i) \log(1 - p)$ .

**Why.**  $\log(uv) = \log u + \log v$  turns the product into a sum, and  $\log(u^c) = c \log u$  brings the exponents down.

**Watch for.**

- “Distributing” the log over a sum, e.g. writing  $\log(1 - p)$  as  $\log 1 - \log p$ .
- Losing the  $(1 - y_i)$  factor.

**3b.**

Suppose the monitoring program had 1,852 nests and each nest contributed a likelihood of about 0.5. Roughly how big is the product of all 1,852 terms? Note that standard computer arithmetic cannot store positive numbers much smaller than  $10^{-308}$ . Why might you want to work with the sum from 3a instead? What is its value here?

**Answer.** The product is  $0.5^{1852} \approx 10^{-558}$ , which a computer rounds to exactly 0. The log sum is  $1852 \log(0.5) \approx -1283.7$ .

**Why.** A product of many numbers below 1 shrinks past what a computer can store (underflow). The sum of logs is an ordinary number.

**Watch for.** Using  $\log_{10}$  for the sum. That gives  $-557.5$ , which is fine as a size estimate but isn't the natural log that R's `log()` and `dbinom(..., log = TRUE)` use.

## Group check-in

**Group question.** Multiplying the four nest likelihoods together depended on an assumption. Name one biological reason it might fail for real sage-grouse nests.

**Answer (any sensible one).** The assumption is independence. It fails when nests share an unmeasured cause:

- the same hen renests after a failure;
- nearby nests share predators (a predator that finds one nest is more likely to find its neighbors);
- nests in the same year share weather;
- nests on the same ridge share habitat quality that cover doesn't capture.

**Why.** Shared causes make outcomes correlated, so the product overstates how much information the data contain.

**Watch for.** Answers about the covariate ("cover differs between nests"). That's a reason  $p$  might differ, not a reason nests are dependent.

## Part 4: Adding the covariate

4a.

Check the predicted  $p$  for nest 2 (20% cover) by hand.

**Answer.**  $a + bx = -2.2 + 0.11(20) = 0$ , so  $p_2 = 1/(1 + e^0) = 0.5$ .

**Watch for.** Sign errors in the exponent. It's  $e^{-(a+bx)}$ , and here  $a + bx = 0$  anyway.

4b.

Assuming the nests are independent, what is the likelihood of this candidate parameter pair,  $\mathcal{L}_4(a = -2.2, b = 0.11)$ ? Careful with the nest that failed.

**Answer.**  $0.75 \times 0.5 \times (1 - 0.25) \times 0.75 = 27/128 \approx 0.211$ . (With unrounded  $p_i$  it's 0.2112.)

**Why.** A nest that hatched contributes  $p_i$ . A nest that failed contributes  $1 - p_i$ .

**Watch for.** Using  $p_3 = 0.25$  for the failed nest instead of  $1 - p_3 = 0.75$ . That gives 0.070, a third of the right answer.

4c.

Compare this to the best of your three constant- $p$  likelihoods from 2c. Which model do the data support more, and by what factor? Does that show that cover really matters for nest success? Give at least one reason for caution.

**Answer.** The covariate model: 27/128 vs. 27/256, a factor of 2. That does not show that cover matters.

**Why (discussion).** Any of these reasons for caution works:

- Four nests is very little data. A factor of 2 could easily arise by chance.
- The covariate model has two parameters and the constant model has one, so it can fit better even if cover is irrelevant.
- $(a, b)$  was handed to us. Nobody searched for the best pair.
- A ratio says nothing yet about uncertainty.

**Watch for.**

- Concluding “cover matters” with no qualification.
- Comparing to  $\mathcal{L}_4(0.5)$  instead of the best constant- $p$  value,  $\mathcal{L}_4(0.75)$ .

4d.

Write the likelihood for  $n$  nests in general,  $\mathcal{L}_n(a, b)$ , using  $p_i = 1/(1 + e^{-(a+bx_i)})$ . Then underline the quantities that are fixed data and circle the quantities that the likelihood is a function of.

**Answer.**

$$\mathcal{L}_n(a, b) = \prod_{i=1}^n p_i^{y_i} (1 - p_i)^{1-y_i}, \quad p_i = \frac{1}{1 + e^{-(a+bx_i)}}$$

Underlined (data):  $y_i, x_i, n$ . Circled (parameters):  $a, b$ .

**Why.** Each  $p_i$  is fixed once  $a, b$ , and  $x_i$  are known, so the only free quantities are  $a$  and  $b$ .

**Watch for.** Circling  $p_i$ . The  $p_i$  are not separate parameters.

## Part 5: From math to R code

Students run this part themselves, with AI discouraged. Expect syntax errors (missing commas, unmatched parentheses). Let them work through the error message before offering a fix.

5a.

Type this function into R, filling in the blanks so it returns the *negative* log-likelihood of a vector of nest outcomes  $y$  for a single success probability  $p$ . `-sum(dbinom(x = _____, size = _____, prob = _____, log = TRUE))`

**Answer.**

```
nll_const <- function(p, y) {
  -sum(dbinom(x = y, size = 1, prob = p, log = TRUE))
}
```

**Why.**  $x$  is the observed data, `size = 1` makes each nest a single trial, and `prob` is the parameter. `dbinom()` returns one log probability per nest, and `sum()` adds them (the sum from 3a).

**Watch for.**

- `x = 1` or `x = p` (confusing data with parameter).
- `size = length(y)`, which treats all nests as one binomial draw.
- Forgetting the minus sign.

5b.

Run `y <- c(1, 1, 0, 1)`, then `exp(-nll_const(0.5, y))`. Which entry of your table in 2c should it match? If it doesn't, use the mismatch to track down the bug.

**Answer.** 0.0625, the  $p = 0.50$  row of 2c (1/16).

**Why.** `exp(-...)` undoes the minus sign and the log, returning the likelihood itself.

**Watch for.** Reporting `nll_const(0.5, y) = 2.77` without converting back. If the number is way off, check `x` and `size` first.

5c.

Run the code below with your own working `nll_const`. Which value of  $p$  gives the smallest negative log-likelihood? Does it agree with your sketch in 2e? Why do you think we work with the *negative* log-likelihood rather than the log-likelihood?

```
p_grid <- seq(0.05, 0.95, by = 0.01)
nll_vals <- sapply(p_grid, function(p) nll_const(p, y))
p_grid[which.min(nll_vals)]
```

```
## [1] 0.75
```

**Answer.**  $p = 0.75$ , matching the peak in 2e.

**Why negative (discussion).** Any of these:

- Optimizers such as `optim()` minimize by default (as with the SSR in Lab 2).
- Discrete likelihoods are at most 1, so their logs are at most 0. Negating gives a nonnegative “badness of fit” score, like SSR.
- Maximizing  $\ell$  and minimizing  $-\ell$  are the same problem.

**Watch for.** Thinking the minimum of the NLL and the maximum of the likelihood could be at different  $p$ . The log is increasing, so they can’t be.

**5d (optional).**

Write the covariate version, using your formula from 4d. If you ran `exp(-nll_cover(c(-2.2, 0.11), y, cover))`, what number should you get, approximately, and why?

**Answer.**

```
nll_cover <- function(par, y, cover) {
  p <- 1 / (1 + exp(-(par[1] + par[2] * cover)))
  -sum(dbinom(y, size = 1, prob = p, log = TRUE))
}
exp(-nll_cover(c(-2.2, 0.11), y, cover))
```

```
## [1] 0.211157
```

About 0.211, the answer to 4b. The small difference from 27/128 comes from rounding the  $p_i$  in 4b.

**Watch for.** Using `par$a` on an unnamed vector, or forgetting the minus sign inside `exp()`.

## Reflection

Now that you’ve written and run your first likelihood function, what part of turning the math into code gave you the most trouble? And more generally, what is still fuzzy about likelihood? What would you like the instructor to review on Monday?

No fixed answer. Read these to decide what to revisit.

- **Common coding trouble:** matching `dbinom()` arguments to data vs. parameter; why `size = 1`; trusting one vectorized call instead of a loop; losing the minus sign.
- **Likely conceptual gaps:** likelihood vs. probability (2d, the most common); why multiply and why log; what “the data are fixed” means; how sure we can be of the peak (uncertainty, which comes later in the semester).